

# Signature of Variable Binding in Episodic Memory

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## Abstract

Brain binding

## 1 Introduction

### 1.1 Reconciling Episodic Memory with Working Memory Solutions

Considering the temporal structure of the task, an alternative hypothesis for task solution is through working memory. Specifically, the task can be reinterpreted as a cued working memory retrieval task, in which presentation of the example sequence serves as a cue for retrieval of the first or the second item in the test sequence. This setting can be mapped directly to a working memory control task where macaques are trained to reproduce one of two simultaneously presented items after a delay, depending on a cue that either precedes or follows stimulus presentation (coined "pro" and "retro" conditions) [1]. For the retro task, macaques learn a solution in which the two stimuli are represented independently in orthogonal PFC neural activity subspaces before selection, and get aligned based on task relevance after cue presentation. In pro trials, alignment is immediate and the subspaces used to represent the selected/unselected items are the same ones used in the retro task. Importantly, the authors establish a correspondence between the transformation of working memory contents used in the two tasks with cross-decoder analyses[1].

Generally, this idea that items are stored in orthogonal subspaces in the working memory and context-dependent transformations are used to route relevant representations to output-potent spaces is a common theme in the working memory literature [2, 3, 4, 1]. A recent theoretical proposal [5].

## 2 Results

### 2.1 Representation Similarity Analysis

## 3 Methods

### 3.1 RSA Analysis

## References

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